

Trend Following: Replication and Out-of-Sample Test

Lempérière, Deremble, Seager, Potters & Bouchaud (2014), “Two centuries of trend following”,
reproduced on free data and tested on the twelve years after publication

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github.com/lwang-genomics/trend-following-replication

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1 Summary

Lempérière et al. (2014, Capital Fund Management) show that a minimal trend-following rule earns significant excess returns across commodities, currencies, stock indices and bonds: a t-statistic of about 5.9 on futures since 1960, and about 10 on spot data back to 1800. This note re-runs their rule on the same 27-market universe using only free public data, checks how the data affect the result, and then asks the question a published result invites: **did it hold after publication?**

- **The result replicates.** For 1960–2013 the aggregate Sharpe ratio is 0.74 (paper: 0.78), with $t = 5.4$ (5.7) and a drift-removed $t^* = 4.5$ (5.0). The trend P&L is only 0.18 correlated with the long-only drift (paper: 0.15). Currencies and bonds match the paper closely. Commodities are much weaker on spot prices (Sharpe 0.17 vs 0.80), as the paper itself anticipates, because spot data miss the futures' carry.
- **Saturation replicates.** A tanh fit of the next move on the signal saturates at $s^* = 1.01$ (paper: 0.89), beats the linear fit ($F = 11.2$), and the cubic term is negative, as reported.
- **After publication the effect is weaker.** Over 2014–2026 the Sharpe ratio is 0.27 ($t = 1.0$), against 0.74 in-sample. A block bootstrap of the in-sample months puts the probability of an out-of-sample Sharpe this low at 9%. So the result is weaker but not statistically rejected, and every n gives the same picture.
- **Two data traps are documented and avoided.** Most free series are **monthly averages**. Trading them the very next month manufactures a fake trend (Working, 1960). The paper's equation, which holds the position one month later, removes the artefact, and on markets with month-end data it reproduces the month-end result exactly (Sharpe 0.22 vs 0.22). **Administered prices** (grain support, oil price controls) need a stale-price filter; without one, a single corn observation from 1972 (-467σ) cuts the 1960–2013 Sharpe ratio from 0.82 to 0.34.
- **150 years of US data** (Shiller, from 1873) give a trend Sharpe of 0.45 with $t = 5.5$, positive in every 50-year block.

2 The paper and what is replicated

The paper defines, for each market and an exponential moving average (EMA) time scale of n months,

$$s_n(t) = \frac{p(t-1) - \langle p \rangle_{n,t-1}}{\sigma_n(t-1)}, \quad Q_n(t) = \sum_{t' < t} \text{sign}[s_n(t')] \frac{p(t'+1) - p(t')}{\sigma_n(t'-1)},$$

where $\langle p \rangle_{n,t}$ is an EMA of past prices excluding $p(t)$ and σ_n is an EMA of absolute monthly price changes. The position is long or short one unit of risk, with no costs. The paper uses monthly closes, $n = 5$, 27 markets (indices, 10-year bonds and currencies of seven countries, plus seven commodities) and Global Financial Data. Its aggregate P&L is the sum of the risk-normalised P&Ls; Sharpe ratios need no risk-free rate because futures are self-financed; $t = \text{Sharpe} \times \sqrt{\text{years}}$; and the “de-biased” t^* is the t-stat of the intercept after regressing the trend P&L on the long-only P&L.

Two implementation details. (i) As printed, eq. (2) pairs the signal built from $p(t'-1)$ with the price change $p(t') \rightarrow p(t'+1)$, so the position is held one month after the information date; this replication follows the equation literally ($\text{lag} = 1$). (ii) An EMA “with decay rate n months” is implemented with weight $\frac{1}{n}$ on the newest observation, and a 24-month warm-up precedes the first position.

3 Data

Sector	Free source (paper: Global Financial Data)	From
Indices (7)	OECD total share price indices (FRED SPASTT01..M661N)	1955–60
Bonds (7)	OECD 10-year yields (FRED IRLTLT01..M156N), turned into a 10-year bond price by exact monthly repricing	1953–69; Japan 1989
Currencies (6)	Fed exchange rates vs USD (FRED); Deutsche mark spliced to the euro in 1999	1973 (end of Bretton Woods)
Commodities (7)	World Bank “Pink Sheet”: maize, US HRW wheat, sugar, beef (live-cattle proxy), copper, US natural gas; WTI crude (FRED); IMF series extend the Pink Sheet	1960; natural gas 1985; crude 1982
US long history	Shiller: S&P composite and 10-year rate	1871; bond 1953

Start dates follow the paper’s rule that only freely traded, moving prices can trend. Currencies start after Bretton Woods; crude starts after US oil price controls ended in 1981 (the price was flat in 63 of 96 months in 1974–81); natural gas starts after wellhead decontrol.

4 Three data checks

4.1 Monthly averages fake a trend unless the position is lagged

The OECD, Fed and World Bank monthly series are averages of daily prices. Averaging a random walk gives its monthly changes a lag-1 autocorrelation of 0.25 (Working, 1960), and every series here shows 0.25–0.35. A rule that trades the month right after the signal harvests that artefact. For the markets where free daily data exist (six currencies and the US 10-year), Table 1 compares averaged and month-end prices directly.

Market	From	AC(1) avg	AC(1) month-end	SR avg, lag 0	SR avg, lag 1	SR month-end, lag 0	SR month-end, lag 1
FX_GBP	1973	0.33	0.06	0.45	0.12	0.22	0.23
FX_JPY	1973	0.32	0.05	0.69	0.37	0.38	0.31
FX_CAD	1973	0.26	−0.06	0.52	0.28	0.08	0.10
FX_CHF	1973	0.26	0.01	0.49	0.06	0.21	0.02
FX_AUD	1973	0.31	0.01	0.45	0.15	0.12	0.09
FX_EUR	1999	0.30	0.04	0.65	0.30	0.30	0.12
BD_US	1962	0.30	0.09	0.58	0.28	0.22	0.10
Mean				0.55	0.22	0.22	0.14

Table 1: Monthly averages vs month-end closes, $n = 5$. AC(1): lag-1 autocorrelation of monthly log changes. Lag 0 trades the next month; lag 1 is eq. (2) as printed.

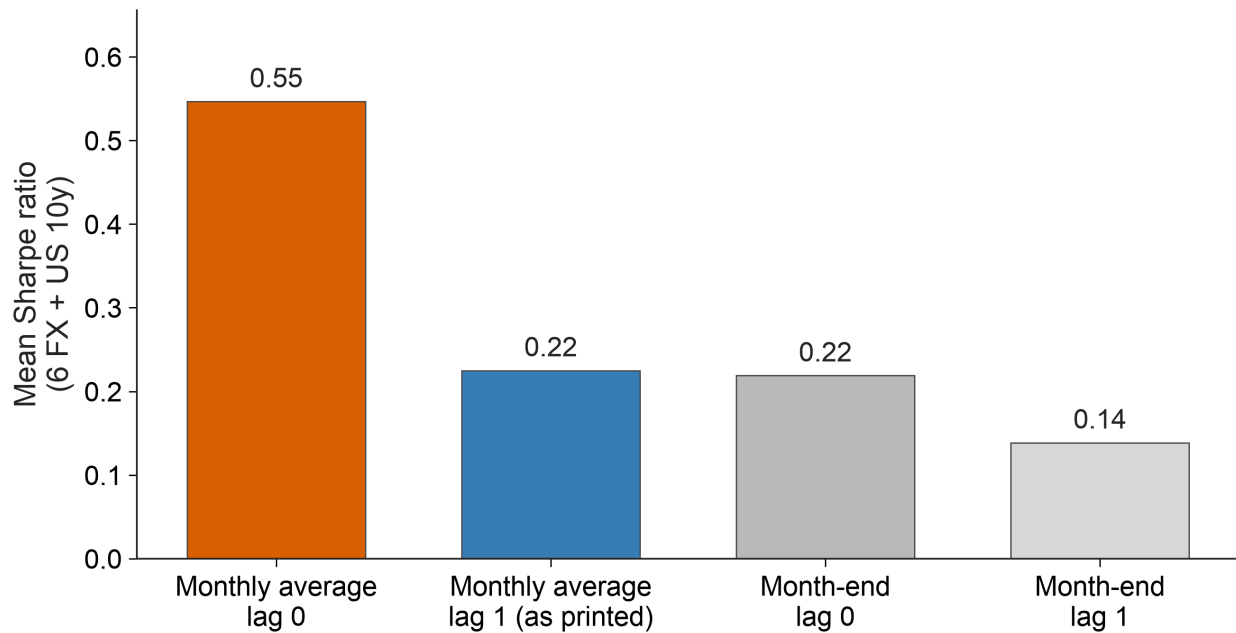


Figure 1: Mean Sharpe ratio over the seven markets in Table 1.

Month-end prices have no autocorrelation (≈ 0.03). On averaged data, lag 0 more than doubles the Sharpe ratio (0.55), a pure artefact; a simulation in the test suite shows that the same rule earns a mean t-stat of about 2 on averaged **random walks**. Lag 1 removes the artefact, and its Sharpe on averaged data (0.22) matches the realistic month-end, lag-0 strategy (0.22). The main specification (averaged data, lag 1) is therefore both faithful to the paper and realistic. The cost is that very short time scales, whose information decays within a month, look weaker than in the paper (Table 2).

4.2 Administered prices need a stale-price filter

Grain prices in the 1960s–70s were set by support programmes and often unchanged for months. σ_n then collapses towards zero, and the first real move is divided by almost nothing: corn in January 1972 contributes -467σ . Without a filter, this single observation pulls the 1960–2013 Sharpe ratio from 0.82 to 0.34. The paper requires “prices actually moving (no gaps)”. Here a market is traded in a month only if its price changed in at least 10 of the previous 12 months. The rule uses only past data, and the result hardly depends on the threshold (Sharpe 0.74, 0.74 and 0.70 for 10, 11 and 12 of 12).

4.3 Interpolated history is not data

Shiller’s long-term rate before 1953 is interpolated from annual observations. Its monthly changes have a lag-1 autocorrelation of 0.92 (0.31 afterwards), which handed the US bond a spurious trend Sharpe of 2.0 in 1873–1899. The long-history bond therefore starts in 1953.

5 Replication, 1960–2013

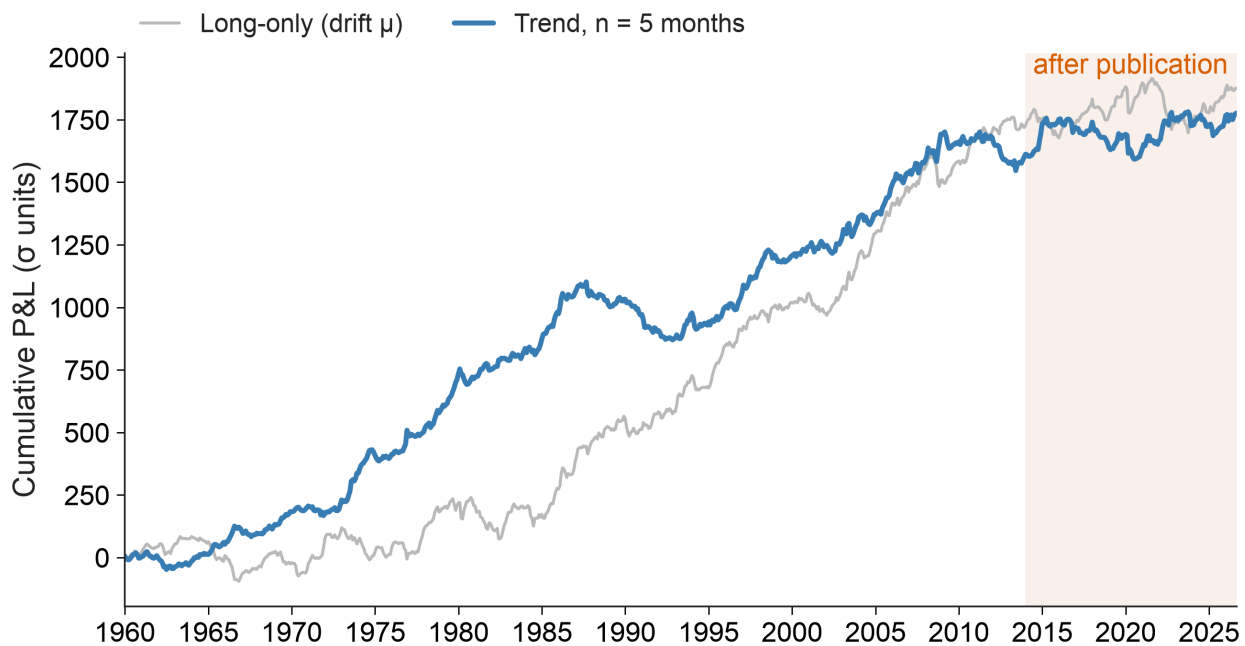


Figure 2: Aggregate trend P&L ($n = 5$, σ units, summed over markets) and the long-only drift. Shaded: after publication.

n (months)	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR 2014–26	t 2014–26
2	0.80	0.39	5.9	2.9	5.5	2.8	0.21	0.7
3	0.83	0.65	6.1	4.8	5.5	4.0	0.13	0.5
5	0.78	0.74	5.7	5.4	5.0	4.5	0.27	1.0
7	0.80	0.70	5.9	5.1	5.0	4.1	0.26	0.9
10	0.76	0.64	5.6	4.7	5.1	3.4	0.30	1.1
15	0.65	0.58	4.8	4.3	4.5	2.4	0.21	0.8
20	0.57	0.53	4.2	3.9	3.3	1.9	0.23	0.8

Table 2: Paper’s Table 1 vs this replication, 1960–2013, plus the out-of-sample Sharpe ratio. t*: drift-removed t-stat.

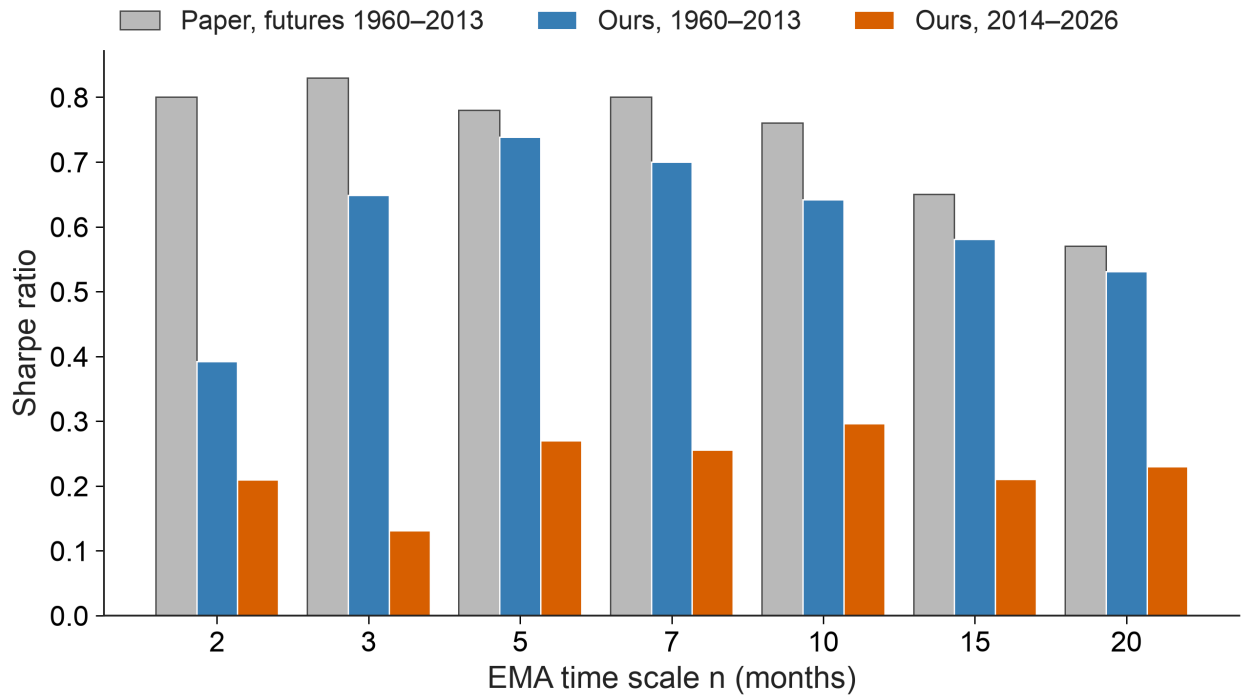


Figure 3: Sharpe ratio by EMA time scale: paper (futures), replication (spot proxies), and after publication.

Sector	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR(μ) ours	Start paper	Start ours	SR 2014–26	t 2014–26
Commodities	0.80	0.17	5.9	1.2	5.0	-0.4	0.69	01/1960	03/1962	0.32	1.2
Currencies	0.57	0.53	3.6	3.3	3.4	3.2	0.16	05/1973	05/1975	-0.02	-0.1
Bonds	0.49	0.45	2.8	3.3	1.6	3.3	0.04	05/1982	01/1960	0.20	0.7
Indices	0.41	0.57	2.3	4.2	2.1	2.6	0.68	01/1982	01/1960	0.13	0.5

Table 3: Paper's Table 2 vs this replication by sector, $n = 5$, through 2013, with the 2014–2026 Sharpe ratio. The paper's bond and index futures start in 1982; the spot proxies start around 1960.

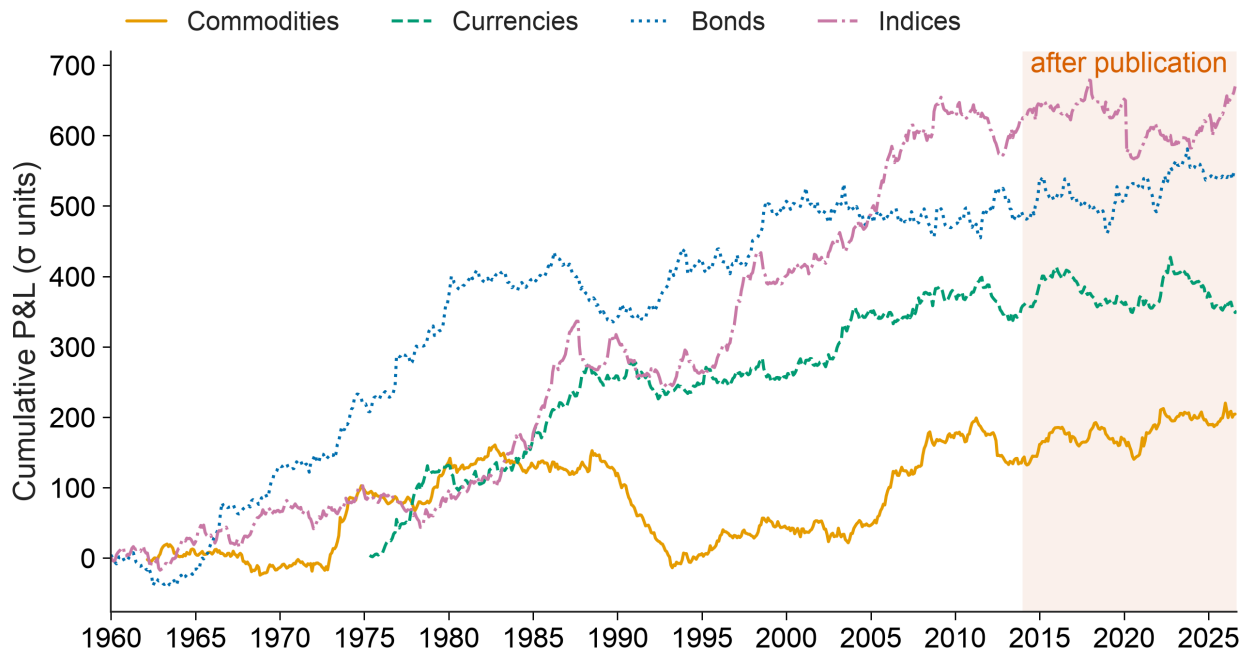


Figure 4: Cumulative trend P&L by sector.

Period	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR(μ) ours	t(μ) ours
1960s	0.66	0.77	2.1	2.5	1.8	2.4	-0.08	-0.3
1970s	1.15	1.46	3.6	4.6	2.5	4.8	0.53	1.7
1980s	1.05	0.79	3.3	2.5	2.9	2.1	0.64	2.0
1990s	1.12	0.42	3.5	1.3	3.0	0.3	1.00	3.2
2000–13	0.75	0.57	2.8	2.1	1.9	1.0	1.00	3.8
2014–26	–	0.27	–	1.0	–	0.8	0.22	0.8

Table 4: Paper’s Table 3 vs this replication by decade, $n = 5$; the last row is after publication.

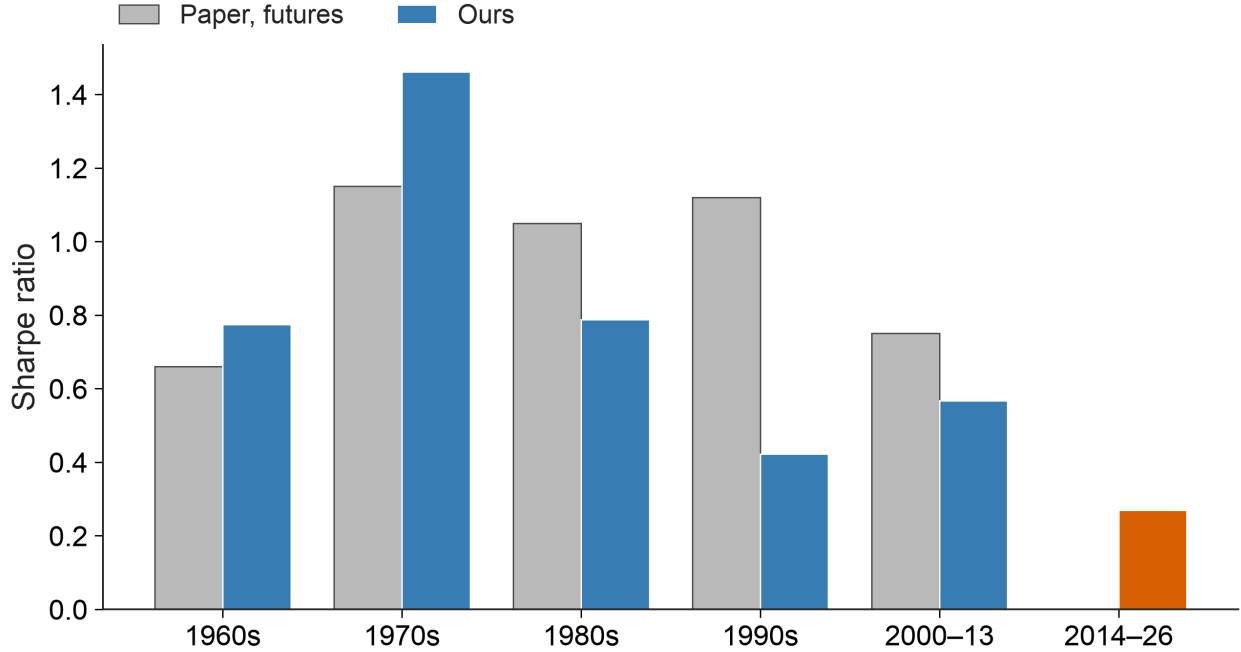


Figure 5: Sharpe ratio by decade. Orange: after publication.

The aggregate replication is close for $n \geq 5$: Sharpe 0.53–0.74 vs 0.57–0.80. Every decade from the 1960s to 2013 is positive. Sector by sector, currencies (0.53 vs 0.57) and bonds (0.45 vs 0.49) match; indices are stronger on the longer spot history. Commodities are the clear gap (0.17 vs 0.80). The paper reports only a 65% spot–futures correlation for commodities, because futures add a carry (roll-yield) term that spot prices lack. Averaged monthly commodity prices, and beef as a stand-in for live cattle, add noise.

6 The signal saturates

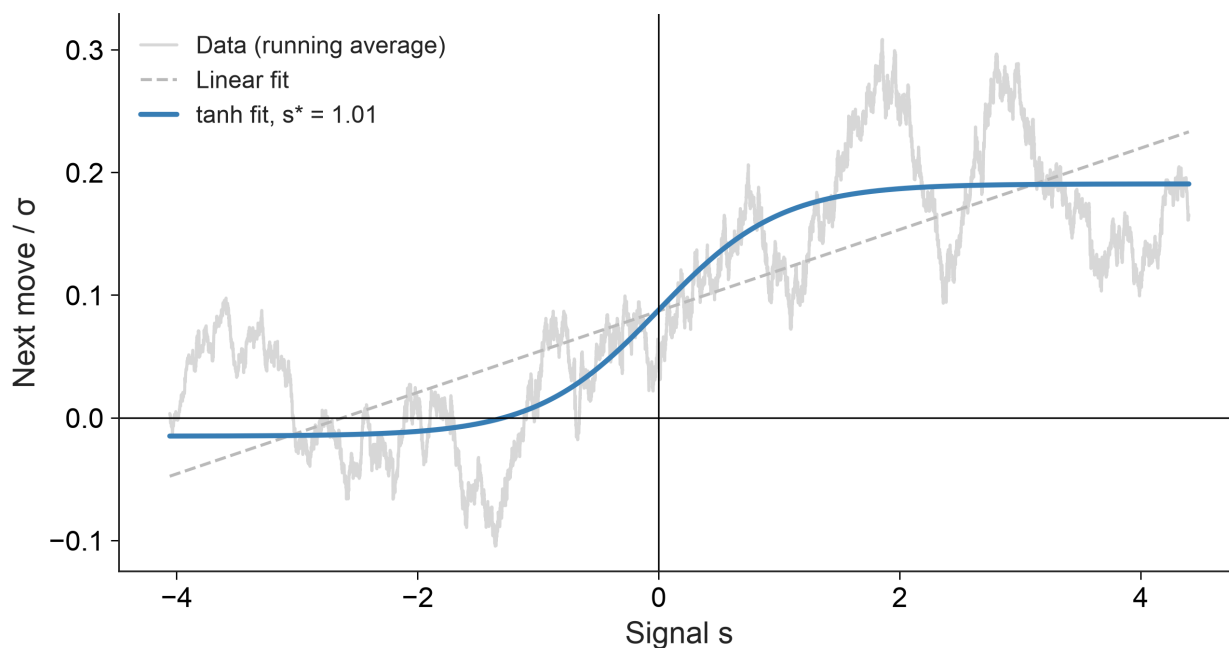


Figure 6: Next normalised move vs signal, pooled over markets ($n = 5$, 18992 points): running average over 1,000 points, linear fit and tanh fit.

Fit	a	b	s^*	Note
Linear, paper	0.018	0.038	–	
Linear, ours	0.087	0.033	–	
tanh, paper	–	0.075	0.89	
tanh, ours	0.088	0.101	1.01 ± 0.61	F vs linear = 11.2

The predictive slope flattens for strong signals. The tanh model is preferred to the linear one, the cubic coefficient is negative (-0.0023), and the saturation scale s^* is close to the paper's. The intercept is larger here because the spot proxies' drift is larger in σ units; the signal is noisy, so s^* is only loosely determined.

7 After publication: 2014–2026

Period	Years	Sharpe	t-stat	Long-only Sharpe
1960–2013 (in sample)	54	0.74	5.4	0.69
2014–2026 (after publication)	12.8	0.27	1.0	0.22

The aggregate Sharpe ratio fell from 0.74 to 0.27 after publication, and 5 of 13 calendar years were negative. Two tests put this in proportion:

- **Difference of Sharpe ratios:** $z = -1.51$, using the standard error $\sqrt{(1/Y_1 + 1/Y_2)}$.
- **Bootstrap:** drawing 12.75-year paths from the in-sample months (12-month blocks), an out-of-sample Sharpe at or below 0.27 occurs 9% of the time; the 90% range is 0.15 to 1.33.

So the post-publication period is unusually weak but not inconsistent with the historical effect. With a Sharpe ratio near 0.7, drawdowns of several years are expected: the paper itself notes that their typical length is about $1/S^2$ years. The weakness is broad, not one sector: indices 0.13, bonds 0.20, currencies -0.02 , commodities 0.32. 2022, the year of the inflation shock, was the best year since 2014. The long-only drift also weakened ($0.69 \rightarrow 0.22$), so the period was poor for simple directional strategies in general.

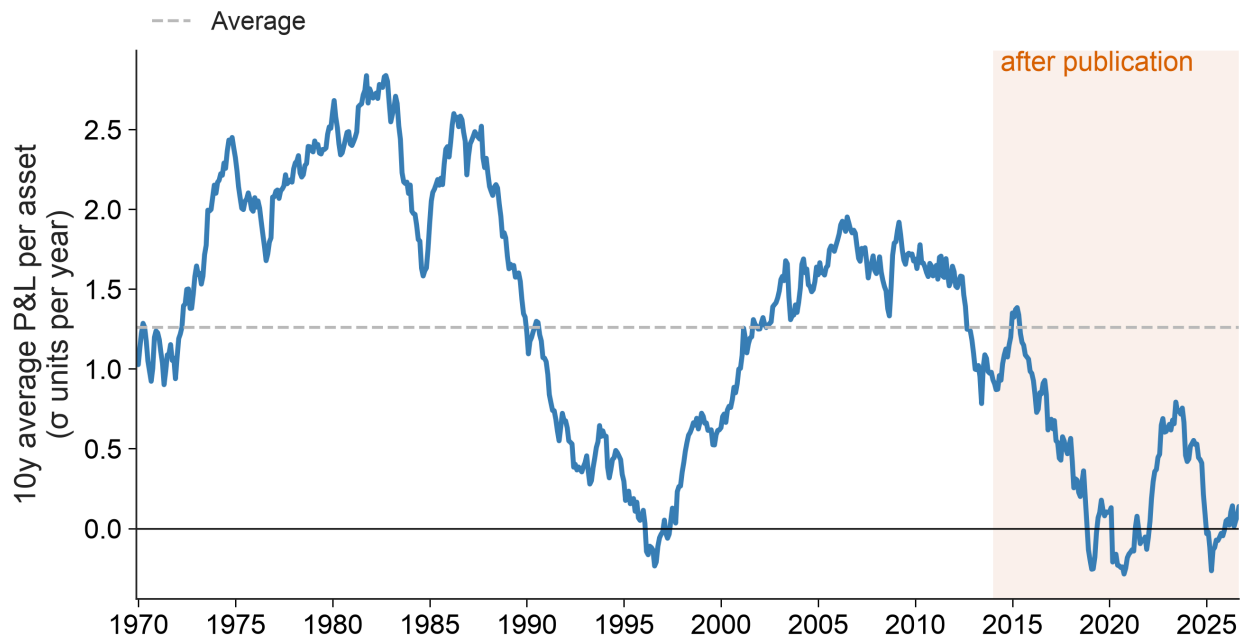


Figure 7: Trailing 10-year average trend P&L per market (σ units per year).

The paper reports that the 10-year performance of the trend was never negative in two centuries. On these free spot proxies it does dip below zero: briefly in 1996 (-0.29 at the low, 2020-10) and again around 2018–2021. So that claim does not replicate on this data, even before publication. The likely cause is the weaker commodity sector, which carried much of the paper's performance.

8 150 years of US data

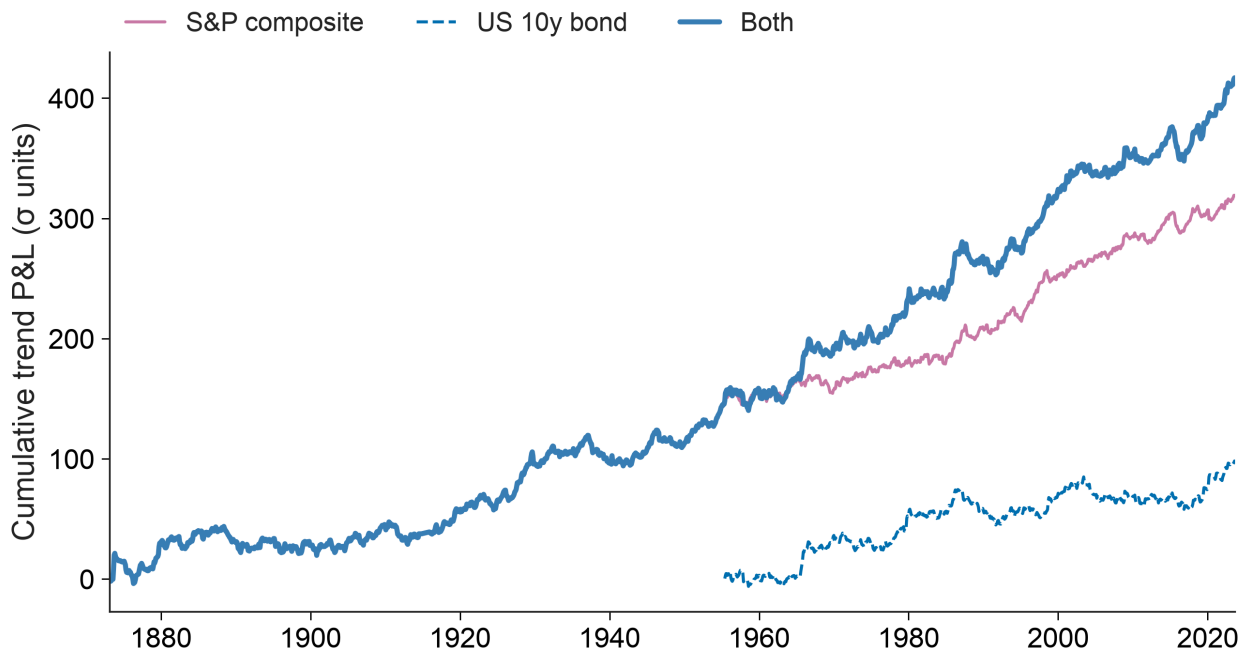


Figure 8: Cumulative trend P&L on the S&P composite (from 1873) and the US 10-year bond (from 1953), Shiller data.

Period	SR (2 assets)	t	t*	SR equity	SR bond
1873–2023	0.45	5.5	5.2	0.42	0.27
1873–1899	0.18	0.9	0.9	0.18	–
1900–1949	0.34	2.4	1.9	0.34	–
1950–1999	0.60	4.3	3.9	0.56	0.30
2000–2023	0.57	2.8	2.3	0.56	0.23

Table 5: Trend on US equity and bonds, Shiller data, $n = 5$, lag 1.

On US data alone, the trend has a Sharpe of 0.45 over 149 years ($t = 5.5$, drift-removed 5.2) and is positive in every 50-year block, consistent with the paper’s long-history finding.

9 Conclusions and limitations

The paper’s central claims replicate on free data: a significant, drift-independent trend effect across asset classes and decades, and saturation of the signal. Its size depends on details the paper does not dwell on. Monthly averages fake a trend unless the position is lagged, administered prices must be filtered, interpolated history must be excluded, and spot commodities miss most of the futures’ trend. After publication the effect weakened to a Sharpe ratio of about 0.27, which is low but within what its own history allows.

Limitations: spot and index proxies instead of futures (no carry; commodities in particular); monthly averages for most series; no transaction costs; a fictitious P&L without portfolio-level risk targeting; and 27 markets, not a full CTA universe.

Next steps. (i) Daily futures and the practitioner’s version of the rule, the moving-average and breakout system with ATR position sizing from A. Clenow’s *Following the Trend*. (ii) Trend following as a convex overlay for a long-only book (Dao et al., 2016, *Tail protection for long investors*), combined with the inverse-volatility study in inverse-vol-futures-overlay.

10 Appendix: code and reproducibility

`uv run trendrep` downloads the raw data once (cached in `data/raw/`, not committed), runs every analysis and writes `results/` and `figures/`. `uv run trendrep-report` compiles this PDF, and `uv run pytest` runs the tests. The tests check the EMA and P&L against hand calculations and verify that there is no look-ahead (changing later prices leaves all earlier signals and P&L unchanged). They also verify that random walks give t-stats centred on zero with unit spread, that monthly averaging fakes a trend at lag 0 but not at lag 1, that trending series are profitable, that stale prices are not traded, and that the saturation fit and the de-biasing recover known parameters.

Reference. Y. Lempérière, C. Deremble, P. Seager, M. Potters, J.-P. Bouchaud (2014), “Two centuries of trend following”, *Journal of Investment Strategies* 3(3); arXiv:1404.3274.